

NIKHILA SREE

+1 430-262-8690 | nikhilasree636@gmail.com | [GitHub](#) | [Portfolio](#)

SUMMARY

AI Engineer with 3+ years of software engineering experience building production GenAI systems, agent workflows, RAG pipelines, model evaluations, and backend services. Shipped AI applications handling thousands of weekly requests, improved retrieval accuracy to 89%, reduced p95 latency by 43%, and automated workflows that previously required hours of manual work.

EXPERIENCE

AI Engineer | Walmart

Sunnyvale, CA | Jul 2025 – Present

- Built an internal AI assistant that investigated catalog, inventory, pricing, and promotion exceptions by retrieving internal policies and querying approved enterprise APIs, supporting **3,500+ requests per week**.
- Increased retrieval accuracy from **76% to 89%** across **1,200 labeled queries** by combining lexical and vector search with metadata filters and reranking.
- Increased end-to-end exception resolution from **68% to 86%** across **6 investigation flows** by preserving case state, selecting APIs based on exception type, retrying transient failures, and routing low-confidence cases to human reviewers.
- Reduced pre-release validation from approximately **1 day to under 2 hours** by automatically replaying labeled cases and blocking regressions in retrieval relevance, API selection, unsupported responses, and response latency.

AI Engineer Intern | SellMate.AI

San Francisco, CA | Jan 2025 – Jun 2025

- Built and productionized a multimodal AI pipeline using OpenAI Vision and AWS Bedrock that converted seller-uploaded product images into validated titles, descriptions, categories, and structured attributes stored in the catalog database.
- Reduced listing creation time from approximately **20 minutes to under 30 seconds per SKU**, a **40x improvement**, by automating image analysis, structured JSON generation, schema validation, retries, fallback handling, and catalog insertion.
- Removed the primary bottleneck in large seller onboarding workflows by replacing sequential catalog processing with parallel file streaming and asynchronous LLM inference without loading complete files into memory.
- Improved the security and maintainability of AI listing workflows by migrating custom sessions to Clerk OAuth 2.0 and introducing versioned, backward-compatible APIs as seller usage increased.

Software Engineer | LTIMindtree

Hyderabad, India | Aug 2021 – Jul 2023

- Developed Java and Spring Boot services that processed **100,000+ billing and transaction records daily**, enforcing **25+ business rules** for payments, duplicate detection, and account eligibility.
- Reduced reconciliation investigation time from **2 hours to 20 minutes** by building a module that compared source files with processed records and generated record-level mismatch reports.
- Cut batch-processing time by **38%** by removing redundant SQL joins, adding targeted indexes, optimizing JDBC batch operations, and eliminating repeated database calls.
- Increased successful batch completion from **94% to 99%** and raised test coverage from **55% to 82%** by implementing checkpoint recovery, idempotent reruns, failure-safe restarts, and **80+ JUnit and Mockito tests**.

TECHNICAL SKILLS

Languages: Python, Java, SQL, TypeScript, JavaScript

Generative AI: LLMs, Agentic AI, RAG, Structured Outputs, Function Calling, Tool Use, Prompt Engineering, Human-in-the-Loop

Frameworks: LangGraph, LangChain, OpenAI APIs, AWS Bedrock, Hugging Face Transformers

Retrieval and Evaluation: Embeddings, Vector Search, Hybrid Retrieval, Metadata Filtering, Reranking, Retrieval Evaluation, Agent Evaluation, Hallucination Detection

Backend and Data: FastAPI, Spring Boot, REST APIs, Microservices, PostgreSQL, Redis, Kafka, Pandas, PySpark

Cloud and Infrastructure: AWS, Kubernetes, Docker, CI/CD, Distributed Tracing, Application Monitoring

Testing: Pytest, JUnit, Mockito, Unit Testing, Integration Testing, API Testing

PROJECTS

[AI Learning-To-Publishing Agent](#) | LangChain, FastAPI, PostgreSQL, Docker, Kubernetes

- Built a multi-step agent platform that transformed technical notes and code examples into structured blog posts, LinkedIn content, and interactive quizzes with retrieval-grounded generation, source validation, and review workflows.
- Deployed the FastAPI and PostgreSQL application with Docker and Kubernetes, adding persistent storage, Secrets, health probes, rolling updates, and **39 passing tests** across agent workflows, authentication, publishing states, and API behavior.

[InterviewBot](#) – Python · LangChain · Neo4j · GPT-3.5 · Streamlit · RAG

- Ingested and chunked **100+ pages** into a Neo4j-backed knowledge graph and implemented top-k retrieval for grounded interview-question responses. Delivered an end-to-end application with FastAPI inference endpoints, LangChain retrieval workflows, and a Streamlit chat interface.

EDUCATION

Northwest Missouri State University — MS, Computer Science — GPA: 4.0/4.0.

Maryville, MO | Aug 2023 – Dec 2024

Jawaharlal Nehru Technological University — B.Tech, CS — GPA: 3.6/4.0.

Hyderabad, India | Jun 2018 – May 2021